Kate J. **Davison**

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PhD student developing a multimodal Graph Transformer Network to predict 3D molecular structures from spectroscopic data. Skilled in Python and data analysis, with experience in machine learning, parallel computing, GPU acceleration, and training models on HPC supercomputers.

Education

Technology Enhanced Chemical Synthesis, University of Bristol
Bristol, UK

PhD, Machine Learning, Computational Chemistry — *Sep 2024 - Current*

- Developing a multimodal Graph Transformer Network for small-molecule structural elucidation by predicting interatomic distances from spectroscopic data. Integrating different spectroscopic modalities to improve 3D molecular structure prediction.

University of Sheffield
Sheffield, UK

MChem, Chemistry — *Oct 2019 - Jun 2023*

- Degree classification: **II:I**

University of East Anglia (UEA)
Norwich, UK

MBBS Medicine — *Oct 2018 - Jun 2019*

Professional Experience

University of Bristol
Bristol, UK

Postgraduate Teaching Assistant — *Sep 2025 - Current*

- Supporting students in developing computational and programming skills
- Explaining technical concepts and helping troubleshoot coding and computational problems

Pharmaxo
Corsham, Bath, UK

Quality Assurance Officer — *May 2024 - August 2024*

- Review batch documentation in line with GMP guidelines.
- Approve final product release in accordance with licensing requirements and GMP.
- Ensure compliance with GMP, COSHH, health and safety regulations, and departmental SOPs.
- Support the QA team with QMS activities, including deviations, investigations, and supplier complaints.

Bristol Royal Infirmary
Bristol, UK

Senior Assistant Technical Officer (SATO) — *Sept 2023 - May 2024*

- Manufactured aseptic products, including eye drops, CIVAS, MABs and lipid syringes, following SOPs and GMP guidelines.
- Transferred products from cleanrooms using validated aseptic techniques.

Computational Skills

PROGRAMMING & SOFTWARE

- **Programming Languages**: Python, Bash, TypeScript, LaTeX
- **Machine Learning**: PyTorch, PyTorch Geometric, scikit-learn
- **Scientific Computing & Data Analysis**: NumPy, SciPy, Pandas, Polars, RDKit, CCDC
- **Data Visualisation**: Matplotlib, Seaborn
- **HPC & GPU Computing**: Isambard, parallel and distributed computing, CUDA
- **Software Development**: Version control with Git, Linux command-line tools
- **AI Tools**: LLM-assisted development and workflow automation
- **Operating Systems**: Linux, macOS, Windows

OPEN-SOURCE SOFTWARE DEVELOPMENT

- **EMS (Efficient Molecular Storage)**: package for processing chemical data and associated properties
- **IMPRESSION**: Graph transformer network for molecular structure elucidation

Publications

1. C. Jones, H. Cooke, **K. Davison**, M. Tilby 2026 *Organic Chemistry Frontiers* A comprehensive guide to unsupervised machine learning for understanding and developing synthetic methodology

Interests

Sport

- Experienced runner, having completed multiple marathons, half marathons and ultramarathons.
- Keen cyclist, currently training for a half Ironman.

Music

- Regularly perform in a symphony orchestra.
- Grade 8 Music Theory, Grade 6 Violin (playing at Grade 8 standard) and Grade 5 Piano (playing at Grade 7 standard).